

DAVID (SEHEON) CHANG

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Education

University of Texas at Austin

Bachelor of Science in Computer Science

Bachelor of Science in Electrical and Computer Engineering

Aug 2023 – May 2028

Austin, Texas

3.6/4.0 Cumulative GPA

Experience

2nd Infantry Combined Division - U.S. 8th Army

Sergeant

Jun 2025 – Dec 2026

Pyeongtaek, South Korea

- Enabled tactical communication across ROK-US combined exercises by providing real-time translation for the United Nations Command, and built an offline ESP32 translator (cosine similarity) for field operations.
- Earned the Army Commendation Medal and Army Achievement Medal, awarded by Major General Lombardo.

Ajou University School of Medicine

Research Intern

Jun 2024 – Sep 2024

Suwon, South Korea

- Published 2 SCI clinical AI papers (11 citations, JMIR JIF 6.0) by building a BERT- and GPT-based NLP pipeline.
- Cut medical-data ETL time by 23% ($O(N \cdot M)$ to $O(N+M)$) by translating R libraries to Python and replacing repeated merges and row-wise lookups with hash joins/maps for OHDSI CDM standardization.

LooxidLabs: CES 2022 Innovation Awards Winning Startup

Paid Project Intern

Jun 2022 – Jan 2023

Seoul, South Korea

- Validated a VR mental-health device by supervising user testing in NYC, recruiting and monitoring 200+ testers.

Projects

Sisyphus | C#, .NET 8, WPF, Windows Registry, Windows Internals (IFEO)

Apr 2026 – Present

- Grew to 50+ users within one week of the V1.0 launch by shipping monthly feedback-driven updates and maintaining active user communication via sisyphusfocus.site.
- Built a focus utility (C#/.NET 8, WPF) that enforces app/website blocks, logs categorized activity, and surfaces AI-generated feedback, with smart reminders (stretch, sleep, idle) and AI-assisted objective & to-do management.

Hearty | Next.js, Python, AWS (Chalice - Lambda, API Gateway / CloudFront / S3), Neon Postgres

Apr 2026 – Present

- Built a senior-activity concierge with a senior-friendly conversational AI navigation UI (Gemini API), delivered as a Next.js SPA on S3 & CloudFront backed by Python AWS Chalice microservices and Neon PostgreSQL.

Tessera | ESP32, C/C++, PN532 NFC, PySide6, AES-256-GCM, local SQLCipher

Mar 2026 – Present

- Led a team to a 7th-of-700 finish in the ROK Army Startup Competition by building an offline EHR dog-tag system.
- Decrypted compressed AES-256-GCM records in under 1s by engineering ESP32 firmware to read over SPI, and built a PySide6/SQLCipher desktop app for offline serial provisioning.

Ajou Course Manager | Django, Docker, React, PDFMiner, OpenAI API, PostgreSQL

Oct 2025 – Feb 2026

- Built and maintain a production course manager at Ajou Univ. Medical School, handling 40+ professors, 200+ students, and 100+ courses via automated CRUD workflows and analytics.

Alpaca Trading Bot | Python, SQLite (WAL Mode), Streamlit, systemd, Telegram API

Jan 2025 – Present

- Generated +15% P&L over 2 months with zero manual intervention by running 3 strategies (RSI, Bollinger, short MA).
- Sustained a 24/7 autonomous trading engine on a Vultr VPS by managing it via systemd unit files, with SQLite Write-Ahead Logging (WAL) telemetry, a Streamlit dashboard, and Telegram alert webhooks.

Campus Involvement / Organizations

Faculty Applicant Interviewer

Chandra Family Department of Electrical and Computer Engineering

Spring 2024

University of Texas at Austin

- Processed 10 faculty hiring interviews for the ECE department; recommended by faculty and the advisory board.

Texas Convergent

IoT Build Team Tech Member

Spring 2024

University of Texas at Austin

- Built an NFC alarm system (disabled only via physical scan) and pitched it to 200+ students and an Austin-based VC.

Directed Reading Program

Natural Language Reasoning Team

Spring 2023 – Spring 2024

University of Texas at Austin

- Engaged in weekly research discussions on CoT and o1 reasoning, advised by UT CS Ph.D. student Zayne Sprague.

Additional Information

Languages: ARM Assembly, C, C++, C#, Java, JavaScript, Python, TypeScript | **Tools/Frameworks:** .NET, AWS, Django, Docker, Git, Kubernetes, Linux, Next.js, Node.js, PostgreSQL, React, SQLite, Streamlit